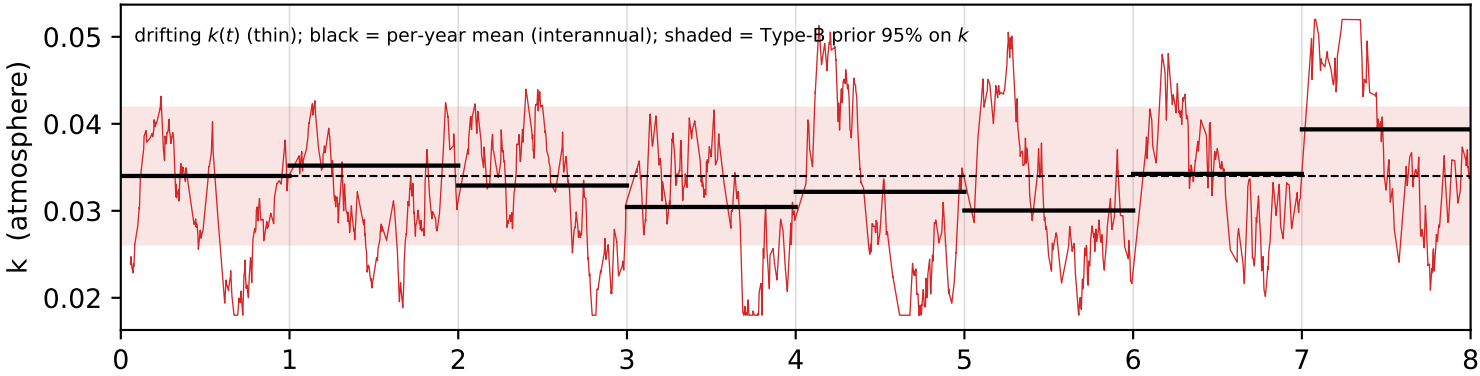
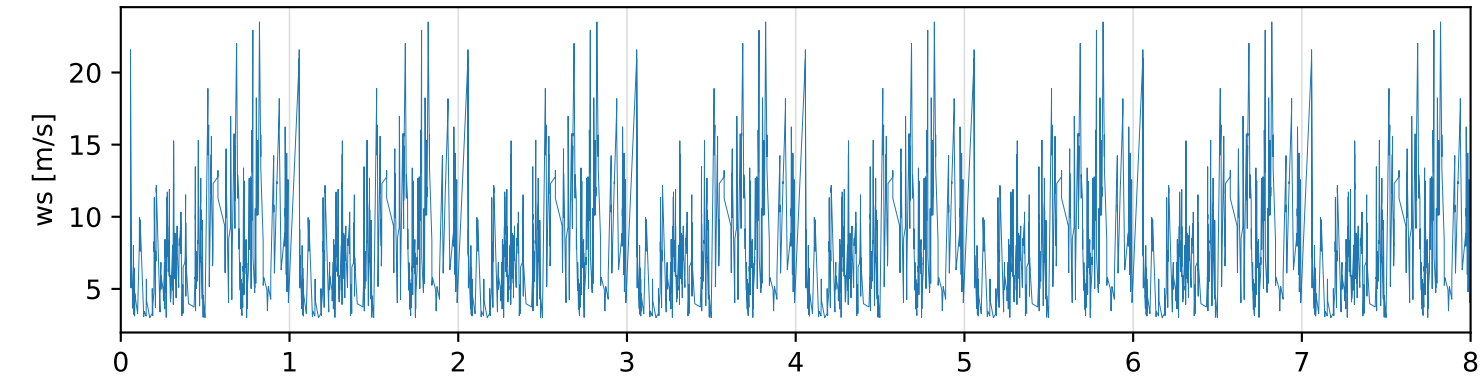
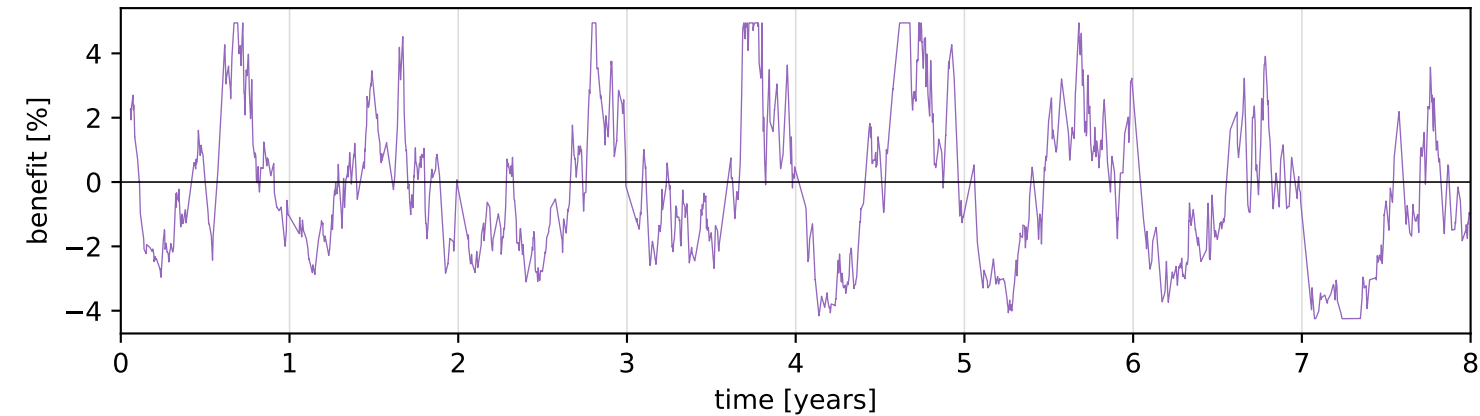


WFFC wake-steering field test — stochastic inputs and Type-B uncertainty

Stochastic time series (8-year campaign, aligned waked sector 266-274°)



Instantaneous true wake-steering benefit drifts and flips sign with k

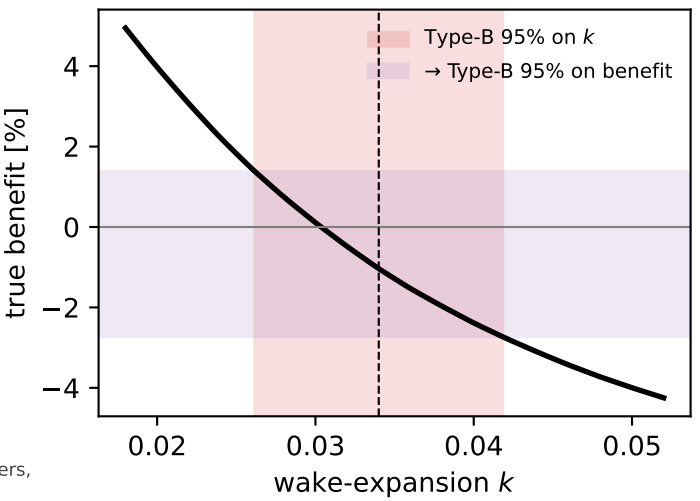


Assumed Type-B uncertainties

input / Type-B	value
epistemic param.	wake expansion k
k prior (Type B)	$N(0.034, 0.004)$
seasonal swing	± 0.008 (aleatoric)
power noise (Type A)	1% / sample
toggle period	70 min, 50/50
control yaw	25° upstream
$d\Delta/dk$	-262 %/unit- k
→ Type-B 1 σ benefit	1.05 %
campaign length	8 years

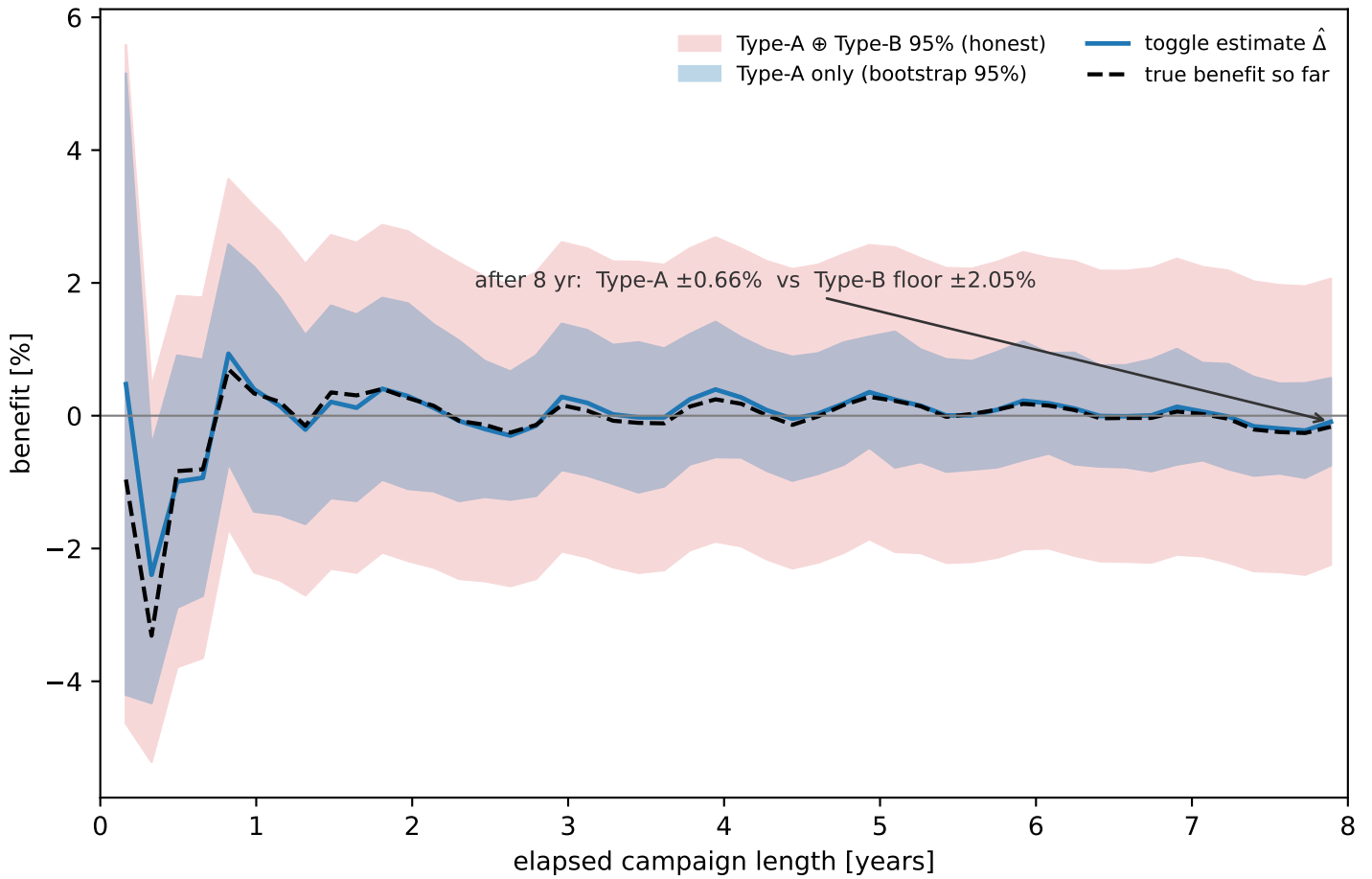
Other field Type-B sources (IEA Task 44): anemometer/vane calibration, yaw-dependent wind-speed/direction transfer functions, power transducers, wake-model form. None are reducible by longer measurement.

Type-B propagation: k uncertainty → benefit



Epistemic mean + uncertainty of the benefit over time windows

Expanding window over 8 years: Type-A (blue) keeps shrinking toward 0, the honest Type-A $\oplus$ Type-B band (red) floors at the epistemic limit



Over many years Type A drops BELOW the Type-B floor: more measurement cannot beat the epistemic limit — 'many measurements are not enough' (rule 3)

